Printed Pages:2 **Sub Code: KCA-055**

Paper Id: 236768 **Roll No.** | | | | | | | | | | |

MCA
(SEM IV) THEORY EXAMINATION 2022-2023
QUANTUM COMPUTING

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a. What are the basic properties of Entropy?
b. What is Block Sphere?
c. Write out the matrix for the three-qubit Toffoli gate.
d. What are the applications of Quantum Computing?
e. What are ion traps?
f. Differentiate between Digital Computer and Quantum Computing.
g. How Quantum information is used in Quantum computing.
h. What is the significance of static and dynamic measures in quantum information?
i. How quantum operations and two types of Quantum Noise.
j. What is Entanglement?

SECTION B

2. Attempt any *three* of the following: 10x3=30

a. Explain the term measurement and unitary evolution with respect to quantum theory concept.
b. Describe the Markov Process for Quantum information.
c. What is the role of data compression in quantum error correction?
d. Explain Qubit and Quantum state. What is the difference between bit and Qubit.
e. What is the purpose of Deutsch Algorithm? How it is separated from Deutsch – Jozsa algorithm.

SECTION C

3. Attempt any *one* part of the following: 10x1=10

a. How the Quantum Search is used for unstructured databases? Explain Grover's search algorithm.
b. Discuss the applications and limitations of Quantum operations.

4. Attempt any *one* part of the following: 10x1=10

a. What are the conditions for Quantum computation? Explain Optical Photon Quantum Computer.
b. How Stabilizer codes are used to compute Quantum Error Correction?

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5. Attempt any *one* part of the following: 10x1=10

a. How fault tolerant quantum computation is done for quantum error correction?
b. Explain the Markov Process for quantum information.

6. Attempt any *one* part of the following: 10x1=10

a. Compare Harmonic Oscillator Quantum Computer and Optical Photon Quantum Computer.
b. Compare DFT and QFT.

7. Attempt any *one* part of the following: 10x1=10

a. How the simulation of Quantum Systems id done, explain with example?
b. Quantum computing helps in speed up the solutions of NP complete problems, illustrate with an example.

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